

Gender Differences in Takotsubo Syndrome.

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Most patients with Takotsubo Syndrome (TTS) are postmenopausal females. TTS in males is rare and gender differences have not been sufficiently investigated. Therefore, we investigated gender differences in TTS. TTS in males and females is often triggered by physical and emotional stress, respectively. Heart failure, a severe in-hospital complication, requires greater mechanical respiratory support in males. Fatal arrhythmias such as ventricular tachycardia and ventricular fibrillation and in-hospital mortality rates are higher in males. The white blood cell (WBC) count has been shown to be higher in males than in females with cardiovascular death compared with non-cardiovascular death. Therefore, the WBC count, a simple marker, may reflect severe TTS. Decreased estrogen levels, common in postmenopausal females, are a pathogenic mechanism of TTS. Females have a more significant increase in the extracellular matrix-receptor interaction than males. Moreover, the pathological findings after hematoxylin-eosin staining were different in males and females. Males had more severe complications than females in the acute phase of TTS; thus, more careful observations and interventions are likely required. From these results, it can be considered that the mechanism of the onset of TTS may be different between males and females. Therefore, it is necessary to fully understand the gender differences in order to more effectively manage TTS.

FOLFOXIRI Plus Bevacizumab as Conversion Therapy for Patients With Initially Unresectable Metastatic Colorectal Cancer: A Systematic Review and Pooled Analysis.

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The combination of fluorouracil, oxaliplatin, and irinotecan plus bevacizumab (FOLFOXIRI-Bev) is an established and effective first-line chemotherapy regimen for metastatic colorectal cancer. However, resection rates of metastases and overall survival with this schedule have never been systematically evaluated in published studies including, but not limited to, the TRIBE (TRIPlet plus BEvacizumab) trial. To assess the clinical efficacy of FOLFOXIRI-Bev, including outcomes and rates of surgical conversions. A systematic review was conducted in October 2016 in concordance with the PRISMA guidelines of PubMed, the Cochrane Central Register of Controlled Trials, SCOPUS, Web of Science, Google Scholar, CINAHL, Ovid, and EMBASE using the terms FOLFOXIRI and bevacizumab and (colorectal cancer). Clinical trials, retrospective case series, and prospective case series that used FOLFOXIRI-Bev for the treatment of initially unresectable metastatic colorectal cancer in humans were included. Individual case reports and retrospective case series with fewer than 10 patients were excluded. Data were extracted independently by 2 reviewers on a predesigned, standardized form. Ultimately, data were aggregated to obtain the pooled effect size of efficacy, according to the random-effects model and weighted for the number of patients included in each trial. Median overall survival and progression-free survival, overall response rates, and rates of R0 surgical conversions and overall surgical conversions. Eleven FOLFOXIRI-Bev studies published between 2010 and 2016 met the inclusion criteria and were pooled for analysis. The studies included 889 patients, with 877 patients clinically evaluable for overall response rates. The objective response rate to FOLFOXIRI-Bev was 69% (95% CI, 65%-72%; I² = 25%). The rate of overall surgical conversions was 39.1% (95% CI, 26.9%-52.8%), and the rate of R0 surgical conversions was 28.1% (95% CI, 18.1%-40.8%). Median pooled overall survival was 30.2 months (95% CI, 26.5-33.7 months) in 6 trials with data available, and progression-free survival was 12.4 months (95% CI, 10.0-14.3 months) in 9 trials with data available. In meta-regression analysis, variables significantly associated with conversion surgery were disease limited to the liver and a higher median number of cycles (close to 12).

For patients with surgically unresectable metastatic colorectal cancer, FOLFOXIRI-Bev is associated with a significant overall response rate. Such an effective regimen leads to a probability of surgical conversion of distant metastases approaching 40%, with more than one-fourth of patients having an R0 resection.

mFOLFOXIRI with or without bevacizumab for conversion therapy of RAS/BRAF/PIK3CA mutant unresectable colorectal liver metastases: the FORBES non-randomized phase II trial.

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The aim of this non-randomized single-center phase II trial was to prospectively assess the clinical efficacy of triplet chemotherapy with modified 5-fluorouracil, folinic acid, oxaliplatin, and irinotecan (mFOLFOXIRI) plus bevacizumab as conversion therapy for initially unresectable RAS/BRAF/PIK3CA mutant colorectal liver-limited metastases (CRLMs). Patients with RAS/BRAF/PIK3CA mutant initially unresectable CRLMs were recruited at a ratio of 2:1 to receive mFOLFOXIRI plus bevacizumab (experimental group) or mFOLFOXIRI alone (control group). The rate of patients attaining no evidence of disease (NED) was the primary endpoint. The secondary endpoints included objective response rate (ORR), depth of tumor response (DpR), secondary resection rate, progression-free survival (PFS), overall survival (OS), and safety. The rate of NED achieved was 40.7% and 30.8%, respectively, in the experimental (n=54) and control groups (n=26); the adjusted odds ratio was 4.519 [95% confidence interval (CI): 1.247-16.375, P=0.022]. The ORR was 77.4% in the experimental group and 60.0% in the control group (P=0.112). The median DpR was significantly greater in the experimental group (45.6% vs. 34.9%, P=0.041). The median PFS was 12.6 months in the experimental group and 9.1 months in the control group [adjusted hazard ratio (HR): 0.584, 95% CI: 0.304-1.121, P=0.106]. Median OS was prolonged in the experimental group compared with the control group (42.6 vs. 35.3 months, adjusted HR: 0.443, 95% CI: 0.195-1.006, P=0.052). Thirty patients (55.6%) in the experimental group and 16 (61.5%) in the control group experienced grade 3/4 adverse events. We observed that the combination of mFOLFOXIRI and bevacizumab increased the rate of clinical NED and showed a trend toward improved survival compared with mFOLFOXIRI alone. This could represent a conversion therapy option for fit patients with initially unresectable RAS/BRAF/PIK3CA mutant CRLMs.

Bevacizumab plus mFOLFOX-6 or FOLFOXIRI in patients with initially unresectable liver metastases from colorectal cancer: the OLIVIA multinational randomised phase II trial.

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For patients with initially unresectable liver metastases from colorectal cancer, chemotherapy can downsize metastases and facilitate secondary resection. We assessed the efficacy of bevacizumab plus modified FOLFOX-6 (5-fluorouracil/folinic acid, oxaliplatin) or FOLFOXIRI (5-fluorouracil/folinic acid, oxaliplatin, irinotecan) in this setting. OLIVIA was a multinational open-label phase II study conducted at 16 centres in Austria, France, Spain, and the UK. Patients with unresectable liver metastases were randomised to bevacizumab (5 mg/kg) plus mFOLFOX-6 [oxaliplatin 85 mg/m², folinic acid 400 mg/m², 5-fluorouracil 400 mg/m² (bolus) then 2400 mg/m² (46-h infusion)] or FOLFOXIRI [oxaliplatin 85 mg/m², irinotecan 165 mg/m², folinic acid 200 mg/m², 5-fluorouracil 3200 mg/m² (46-h

infusion)] every 2 weeks. Unresectability was defined as ≥ 1 of the following criteria: no possibility of upfront R0/R1 resection of all lesions; $<30\%$ residual liver volume after resection; metastases in contact with major vessels of the remnant liver. Resectability was evaluated by multidisciplinary review. The primary end point was overall resection rate (R0/R1/R2). Efficacy end points were analysed by intention-to-treat analysis. In patients assigned to bevacizumab-FOLFOXIRI ($n = 41$) or bevacizumab-mFOLFOX-6 ($n = 39$), the overall resection rate was 61% [95% confidence interval (CI) 45% to 76%] and 49% (95% CI 32% to 65%), respectively (difference 12% ; 95% CI -11% to 36%). R0 resection rates were 49% and 23% , respectively. Overall tumour response rates were 81% (95% CI 65% to 91%) with bevacizumab-FOLFOXIRI and 62% (95% CI 45% to 77%) with bevacizumab-mFOLFOX-6. Median progression-free survival (PFS) was 18.6 (95% CI 12.9 - 22.3) months and 11.5 (95% CI 9.6 - 13.6) months, respectively. The most common grade 3-5 adverse events were neutropenia (bevacizumab-FOLFOXIRI, 50% ; bevacizumab-mFOLFOX-6, 35%) and diarrhoea (30% and 14% , respectively). Bevacizumab-FOLFOXIRI was associated with higher response and resection rates and prolonged PFS versus bevacizumab-mFOLFOX-6 in patients with initially unresectable liver metastases from colorectal cancer. Toxicity was increased but manageable with bevacizumab-FOLFOXIRI. NCT00778102.

Update on the optimal management of patients with colorectal liver metastases.

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Patients with colorectal liver metastases represent a distinct subset of metastatic colorectal cancer. Optimal management requires a multidisciplinary approach, local and systemic. Curative hepatic surgery is standard for resectable cases, but unfortunately, the majority of patients are not initially resectable due to the size, location, and/or extent of disease, inadequate remnant liver volume, or comorbidities. Other local approaches may be complementary (such as portal vein embolization) or alternative (such as ablation, hepatic arterial infusion, selective internal radiation therapy, and stereotactic body radiotherapy) to surgery. Systemic therapy can downsize disease, allowing surgical resection and, potentially, long-term survival, but it must be balanced against the potential for hepatotoxicity. Current standard approaches including cytotoxics and biologics, such as bevacizumab and particularly anti-epidermal growth factor receptor therapy, improve response rates and may enhance downsizing and resection rates. Optimization of local therapies and systemic conversion strategies via controlled, randomized trials is still a pending question.

Chemotherapy in resectable or potentially resectable colon cancer with liver metastases.

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The treatment of colorectal cancer liver metastases has seen significant improvement in recent years and, for certain patients, the long-term survival and even cure are possible. Despite this improvement, many more questions are yet to be answered: the optimal combination, duration, sequence of therapies, role of biologics and the timing of surgical resection are debated in the literature, with conflicting trial results. In this review, the authors highlight the current trial evidence for systemic chemotherapy and biologic therapy for colorectal cancer liver metastases in both the pre and post-resection setting. The treatment of colorectal liver metastases requires a multidisciplinary approach. The role of adjuvant

chemotherapy with 5 fluorouracil and oxaliplatin in stage 3 colon cancer is well established. However, the options for patients with resectable or borderline liver metastases, either in the neoadjuvant or adjuvant settings, require further study. For patients with borderline resectable metastases, the combination of triplet chemotherapy with 5 fluorouracil, oxaliplatin and irinotecan (FOLFOXIRI) offers the best conversion rate. The role of biologic agents such as bevacizumab and EGFR inhibitors in these settings is less clear based on current evidence.

[Rethinking of the conversion therapy in colorectal cancer patients with liver metastasis].

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The incidence of colorectal cancer liver metastasis (CRLM) has gradually increased in recent years. Surgical resection is the main method to achieve long-term survival for patients with CRLM. However, only 20% of these patients have the chance to undergo surgical resection. If the unresectable metastases can be converted to resectable ones by effective conversion therapy, the 5-year survival rate of patients received liver resection can exceed to 30%, which is significantly better than palliative treatment. Therefore, for patients who are initially unresectable, rationally developing a conversion therapy strategy to convert the initial unresectable CRLM into resectable ones is the key to improve the long-term survival of CRLM patients. However, there are still many controversies in clinical practice. In this article, we discuss three critical issues related to the conversion therapy for CRLM based on previous related researches and our experience, including the applicable population of conversion therapy, how to choose a conversion regime and the recognition and treatment of disappeared lesions after chemotherapy.

Potentially resectable metastatic colorectal cancer: an individualized approach to conversion therapy.

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Colorectal cancer is one of the most common cancers worldwide. In recent years, the survival of patients with metastatic disease has improved due to the developments in both medical and surgical care. Patients with technically unresectable metastatic disease could benefit from a multidisciplinary approach for their possible shift toward a technically resectable condition; the choice of the most effective systemic treatment is then crucial to allow conversion to resectability. Systemic conversion therapy may include chemotherapy agents' combinations (fluoropyrimidine, irinotecan and oxaliplatin), with or without targeted agents (cetuximab, panitumumab, bevacizumab). The choice of the best treatment option has to be evaluated by taking into account each patient's baseline characteristics, biological and pathological information and surgical strategy. In particular, the role of some biologic characteristics of the disease, namely the mutational status of EGFR-pathway oncogenes, is emerging as an important predictive factor of response to anti-EGFR targeted agents. Patients presenting with colorectal cancer metastases should be evaluated for multimodal management with curative intent as the appropriate chemotherapy regimen may induce tumor shrinkage, conversion to resectability and improved survival.

Conversion strategies with chemotherapy plus targeted agents for colorectal cancer liver-only metastases: A systematic review.

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There is no consensus on the optimal systemic conversion therapy in patients with unresectable colorectal cancer liver-only metastases (CRLM) to achieve a complete resection. Interpretation of trials is complicated by heterogeneity of patients caused by emerging prognostic and predictive characteristics, such as RAS/BRAF mutation status, lack of consensus on unresectability criteria and lack of data on clinical outcome of secondary resections. A systematic review was performed of characteristics of study populations and methodology of trials regarding patients with initially unresectable colorectal cancer liver-only metastases. Phase II/III randomised trials, published after 2008, regarding first-line systemic conversion therapy in patients or subgroups of patients with CRLM were included. Data on secondary resection outcomes were collected. Overall, 20 trials were included for analysis: seven prospective trials in patients with unresectable CRLM and 13 trials in the overall population of unresectable metastatic colorectal cancer (mCRC) with retrospective subgroup analysis of CRLM patients. Fourteen trials did not provide unresectability criteria at baseline, and criteria differed among the remaining studies. Trials and study populations were heterogeneous in prognostic/predictive factors, use of primary end-points, and reporting on long-term clinical outcomes. R0-resection rates in CRLM patients varied between CRLM studies and mCRC studies, with rates of 22-57% and 11-38%, respectively. Cross-study comparison of (subgroups of) studies regarding first-line systemic treatment in patients with unresectable CRLM is hampered by heterogeneity in study populations, trial designs, use of (K)RAS/BRAF mutational tumour status, and differences/absence of unresectability criteria. No optimal conversion systemic regimen can be selected from available data. Prospective studies with well-defined criteria of these issues are warranted.

Conversion chemotherapy using cetuximab plus FOLFIRI followed by bevacizumab plus mFOLFOX6 in patients with unresectable liver metastases from colorectal cancer.

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Recently, in patients with unresectable colorectal liver metastasis, liver resection sometimes becomes possible by intensive systemic chemotherapy, i.e. conversion therapy. However, among cases that do not respond well to first-line chemotherapy, it is rare that second-line chemotherapy results in a marked response allowing liver resection. We consider that the liver resection rate may be increased by initiating second-line treatment at an earlier stage before progression subsequent to first-line chemotherapy. We are conducting a multicentre Phase II study to evaluate the efficacy and safety of sequential chemotherapy using six cycles of cetuximab plus FOLFIRI (5-fluorouracil, folinic acid and irinotecan) followed by six cycles of bevacizumab plus FOLFOX (5-fluorouracil, folinic acid and oxaliplatin) as conversion chemotherapy. The primary endpoint is the liver resection rate during the bevacizumab + FOLFOX phase. Fifty patients are required for this study.